Database for storing and processing information regarding the Airbnb use case

Finalization phase

DLBDSPBDM01 – Project: Build a Data Mart in SQL

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1. Introduction

This project is centred on designing and implementing a relational database system for an Airbnb platform. Its primary objective to efficiently manage guest transactions and bookings, while the administrative side focuses on improving the service quality and standards. Through consistent and organized data flow, the system aims to attract more customers and increase overall income. To build a well-structured database, the project followed three main phases: conception, development and finalization. An ERD containing twenty-one entities was created during the conception phase. Each table was structured using appropriate primary keys, unique indexes and foreign key constraints. This framework worked as a foundation for the development phase.

In the implementation stage, MySQL workbench was used to create tables and insert test data. We incorporated constraints such as NOT NULL, UNIQUE, and cascading actions to enforce rules and ensure the integrity of relationships between tables. All tables were populated with dummy data, each with twenty entries or fewer. To verify functionality, SELECT and JOIN queries were executed to confirm that relationships between tables worked as expected.

In the final stage of the project, I assessed the database to ensure that all processes were running smoothly. The system was able to perform several operations without complications, including allowing guests to make bookings, tracking payments, associating each listing with its assigned host and storing all records of customer feedback after completed stays.

Some tables such as regulations and security features were difficult to fill because dummy data was limited. Linking foreign keys across different tables was also complex since each table had unique attributes. However, these challenges helped me understand the significance of choosing the right constraints as doing so ensures that queries execute properly. MySQL workbench also became easier to use and exploring more of its tools helped me learn how to fix errors on my own. Overall, the project enhanced my knowledge on how to solve real implementation problems and provided insight on how a database is built practically.

2. Database schema overview

The following table gives us an overview of the stored metadata in the database, including the number of rows and estimated data size:

No.	Name	Rows	Avg_row_length	Data_length	Index_length
1	address	20	819	16384	0
2	admin	21	780	16384	16384
3	amenities	20	819	16384	0
4	availability	20	819	16384	0
5	booking	20	819	16384	0
6	guest	20	819	16384	16384
7	host	20	819	16384	16384
8	housekeeper	20	819	16384	16384
9	insurance	15	1092	16384	0
10	listing	11	1489	16384	32768
11	location	20	819	16384	0
12	payment	20	819	16384	32768
13	pets	10	1638	16384	0
14	property features	9	1820	16384	0
15	property pricing	9	1820	16384	16384
16	rating	13	1260	16384	16384
17	region	15	1092	16384	16384
18	regulations	7	2340	16384	16384
19	reviews	10	1638	16384	32768
20	security features	10	1638	16384	0
21	service provider	21	780	16384	16384

The database is fully documented with the metadata table giving us a clear summary of all the data stored within it.